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B.Tech. Degree VII Semester Examination November 2017

ME 1701 REFRIGERATION AND AIR CONDITIONING (2012 Scheme)

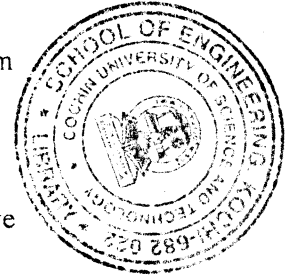
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define one tonne of refrigeration.
- (b) Describe briefly the working of Pulse tube refrigeration with a neat figure.
- (c) In a vapour absorption refrigeration, heat is supplied to the generator at a temperature of 90°C. The cooling in condenser and evaporator takes place at 20°C and -10°C respectively. Find the maximum COP of the system.
- (d) List the differences between vapour compression and vapour absorption refrigeration system.
- (e) Explain the working of an axial flow compressor. How does it differ from a centrifugal compressor?
- (f) Describe briefly the vane type compressor with a neat sketch.
- (g) Write short notes on by-pass factor for cooling coils.
- (h) What is 'effective temperature'? What are the factors governing effective temperature?



PART B

(4 × 15 = 60)

- II. (a) Ice is formed at 0°C from water at 20°C. The temperature of the brine is -8°C. Find (i) COP (ii) kilogram of ice formed per kWh. Assume the cycle is reversed Carnot cycle. Take latent heat of ice as 335 kJ/kg. (8)
 - (b) In a Bell-Coleman cycle air refrigerator, air is compressed to 5 bar from 1 bar and its initial temperature is 10°C. After compression, the air is cooled upto 20°C in a cooler. Calculate COP and net refrigerating effect. Take $C_p = 1.005 \text{ kJ/kgK}$ and $C_v = 0.718 \text{ kJ/kgK}$. (7)
- OR**
- III. (a) What is meant by thermo electric refrigeration? Explain. (5)
 - (b) With the help of p-v and T-s diagrams, explain the working of Bell-Coleman cycle air refrigerator. (10)

(P.T.O.)

- IV. A Freon – 12 vapour compression refrigeration system has a condensing temperature of 50°C and evaporating temperature of 0°C. The refrigeration capacity is 7 tonnes. The liquid leaving the condenser is saturated liquid and compression is isentropic. Determine : (15)
- Mass flow rate
 - Power required to run the compressor.
 - COP of the system.

The properties of Freon – 12 are

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Enthalpy (kJ/kgK)	
		Liquid h_f	Vapour h_g	Liquid S_f	Vapour S_g
50°C	12.199	84.868	206.98	0.3034	0.6792
0°C	3.086	36.022	187.397	0.1418	0.6960

OR

- V. (a) Briefly explain the practical ammonia-water absorption refrigeration system with a neat figure. (9)
- (b) Write short note on multi pressure refrigeration systems. (6)
- VI. (a) In a two stage reciprocating compressor, derive an expression of intercooler pressure (p_2) as the geometric mean of suction pressure (p_1) and condenser pressure (p_3). (9)
- (b) Define volumetric efficiency of a reciprocating compressor and also derive an expression for volumetric efficiency. (6)

OR

- VII. Explain briefly the types of (i) condensers and (ii) evaporators used in refrigeration systems with neat sketches. (15)
- VIII. (a) Differentiate between summer and winter air conditioning systems. (5)
- (b) In a laboratory test, a psychrometer recorded dry bulb temperature as 35°C and wet bulb temperature as 28 °C. Calculate : (10)
- Vapour pressure.
 - Relative humidity.
 - Specific humidity.
 - Dew point temperature.
 - Enthalpy.

OR

- IX. (a) What are the procedures for cooling load calculation of an air conditioned system? (5)
- (b) Explain duct design methods and air distribution arrangements in air conditioning systems. (10)
